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Jiashen Liu

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Research Interests

My research focuses on **embodied intelligence and real-to-sim-to-real (R2S2R) systems**, with an emphasis on reconstructing and adapting realistic simulation environments for embodied agents. I am interested in 3D computer vision and geometric perception, simulation-driven learning and transfer for robotics, and multimodal and LLM-based agents that perceive, reason, and act in complex environments.

Education

University of Warwick, Coventry, UK 2024.09 – 2025.09
Cyber Security Engineering, Master of Science (with **Distinction**)

Nanjing Agricultural University, Nanjing, China 2020.09 – 2024.06
Information Management and Information Systems, Bachelor of Management

- **Average Grade:** 84.25/100
- **Graduation Thesis:** "DiaKG-GPT: An Intelligent Medical Q&A System via Data Augmentation" (Outstanding Graduation Thesis Award)

The Chinese University of Hong Kong, Hong Kong, China 2023.07 – 2023.08
Summer Program

- Completed Technology, Science and Environment courses spanning innovation, sustainability, biology, and Python data analysis; received a government scholarship for overseas study following an excellent final project report

Selected Publications and Manuscripts

Published / Accepted

Li, C., Huang, S., **Liu, J.**, Lei, M., Xing, J., Wu, K., Wang, L., & Luo, F. (2026). FED-FSTQ: Fisher-Guided Token Quantization for Communication-Efficient Federated Fine-Tuning of LLMs on Edge Devices. *(Accepted, IEEE TMC)* 2026

Liu, X., Liu, B., Liu, Y., **Liu, J.**, & Wang, A. (2026). Active Learning-Driven Ensemble Framework for Efficient and Robust Reliability Analysis in Geotechnical Engineering. *(Machine Learning and Data Science in Geotechnics, 2(1), 55–66)* 2026

Yu, X[†], **Liu, J.**[†], et al. (2026). SynUW-VIBench: A Synthetic Underwater Visual-Inertial Benchmark with Frame and Event Cameras. *(Oral Presentation, ICARCV 2026)* 2026

[†] *Equal contribution*

Preprints / Under Review

Liu, J., Zhou, J., Zhang, K., Yu, X., & Hong, Z.* (2026). CAVERN: Controllable and Validated Environments for Robotic Navigation in Underwater Caves. *(Submitted to ICRA 2027)* 2026

Li, C., Wang, L., Lei, M., **Liu, J.**, Zhang, Y., Wu, K., & Luo, F. (2026). PI-TTA: Physics-Informed Source-Free Test-Time Adaptation for Robust Human Activity Recognition on Mobile Devices. *(Preprint; submitted to IEEE TMC)* 2026

Manuscripts in Preparation

Liu, J., Xia, S. (2026). How Do VLMs Actually See Video? A Geometric Account of Visual Context Representations in Video–LLMs. *(Manuscript in preparation for ICLR 2027)* 2026

Research Experience

Embodied Intelligence and Real-to-Sim-to-Real Simulation Reconstruction 2026.10 – Present

Research Assistant, PAIRS Lab, HKUST (Guangzhou)

- Conducting research on embodied intelligence, focusing on real-to-sim-to-real (R2S2R) simulation reconstruction
- Studying reconstruction and simulation workflows to support the transfer of embodied systems between real and simulated environments

Underwater Cave 3D Reconstruction and Simulation-Driven Adaptation 2026.07 – 2026.09

Research Intern, Harbin Institute of Technology Shenzhen (HITSZ)

- Built a multimodal underwater-cave data-acquisition pipeline, calibrating AC1 and Insight9 sensors, synchronizing sensor streams, and reconstructing cave scans with Agisoft Metashape
- Evaluated feed-forward 3D reconstruction models, including VGGT, and identified failure modes caused by low illumination, severe color attenuation, backscatter, turbidity, specular surfaces, and sparse or repetitive textures
- Designing a simulation-driven adaptation pipeline that scans geometrically similar terrestrial caves, reconstructs high-fidelity digital twins, and imports them into HoloOcean and Stonefish
- Proposing controllable underwater multi-view data generation with accurate geometric supervision to adapt VGGT-like models and build reusable cave environments for robotic perception, navigation, planning, and policy training

WebPentestRAG: Retrieval-Augmented Generation for Automated Web Application Penetration Testing 2024.09 – 2025.09

Master's Dissertation Project, University of Warwick

- Designed a RAG-enhanced framework that combines structured reasoning with dynamic knowledge retrieval, improving performance by 42% over baseline LLMs (GPT-4o: 88% task success; Qwen2.5-72B: 72%) in automated penetration-testing tasks
- Architected a three-module system: a RAG-enhanced reasoning module with a Web Task Graph (WTG) for state management, a generation module with two-stage grounded generation and dry-run validation, and a planning agent for workflow orchestration
- Built a specialized security knowledge base of 5,000+ documents from CVE/NVD, Exploit-DB, and OWASP; integrated BGE-M3 (1,024-dimensional) embeddings with a Milvus vector database using HNSW indexing
- Evaluated multiple LLMs using a human-in-the-loop protocol and ablation studies, quantifying the contribution of each component

MIKRAG: Knowledge-Enhanced Multi-Intent Retrieval with Adaptive Multimodal Fusion 2025.03 – 2025.09

- Designed and implemented a large-scale multimodal PDF retrieval system using ColQwen2.5 and BGE embedding models, improving document-search accuracy by 23.7% through query-intent decomposition and adaptive modality weighting
- Built an end-to-end pipeline for PDF processing, text–image embedding, Vespa vector indexing, and dynamic query

optimization; extended it with OmniThink-guided search-space expansion to identify information gaps in initial results

- Applied binary quantization to compress ColPali-generated vectors to 1/32 of their original size while preserving retrieval quality, achieving 4× faster retrieval
- Developed token/patch/document-level fusion and multi-intent branching to improve coverage, precision, and recall

DiaKG-GPT: An Intelligent Medical Q&A System via Data Augmentation 2024.05

- Developed DiaKG-GPT, a data-augmented LLM system for diabetes Q&A; created the DiaMediQA dataset by integrating real-world Q&A, ChatGPT-optimized data, and the DiaKG knowledge graph
- Fine-tuned Baichuan2-7B with LoRA/SFT and RLAIF; used GPT-4-based automatic evaluation to validate improvements in response accuracy and depth
- Received Outstanding Graduation Thesis Award for the undergraduate thesis based on this project

Competitions and Projects

GeoMind · **OreSense** — Global Excellence Award, Entrepreneur Dreamer Competition 2026.04

Team Leader · Hosted by Xi'an Jiaotong-Liverpool University (XJTLU)

- **Led a cross-disciplinary team** to design **GeoMind** · **OreSense**, an AI + digital-twin platform for early-stage critical-mineral exploration, projected to cut decision cost by **40**× while reaching **60%+** success rate; designed the Self-Evolving GeoAgent and a consulting → SaaS → data-ecosystem roadmap, then delivered the English pitch that won the Global Excellence Award

Professional Experience

AI Algorithm Engineer (Part-time), Guangdong Zhongsuanyun Co., Ltd. – Guangdong, China 2025.08 – 2025.12

- Architected a hierarchical multi-agent AI system with one- and two-level agent classifications, supporting 3,000+ concurrent requests for enterprise sales-training and customer-service platforms
- Built LLM-powered conversational agents with LlamaIndex and QwenPlus, using ReAct-style reasoning and tool integration across 20+ specialized agent types
- Implemented a RAG-based customer-service platform with mem0 persistent memory, MongoDB/Redis-backed microservices, and distributed orchestration; achieved 0.92+ similarity matching for personalized conversations

Research Assistant, University of Warwick – Coventry, UK 2024.10 – 2025.01

- **Geological strength prediction:** developed spatial prediction models using Kriging interpolation with prior information; built a Python framework integrating spatial correlation functions and prior knowledge, with coordinate standardization for numerical stability; evaluated via R^2 / RMSE and spatial distribution maps
- **Geological slip-distance classification:** developed multiclass models with LightGBM, CatBoost, and Gradient Boosting; addressed class imbalance via SMOTE; validated robustness through confusion matrices and comprehensive evaluation frameworks

Skills

Agentic AI: Retrieval-Augmented Generation, ReAct and Tool Use, Multi-Agent Orchestration, Persistent Memory

Machine Learning: LoRA/SFT, RLAIF, Multimodal Embeddings, Vector Search, Binary Quantization

Programming & Systems: Python, C/C++, SQL; PyTorch, Hugging Face, LlamaIndex; Milvus, Vespa, MongoDB, Redis, MySQL

Languages: Mandarin (Native), Cantonese (Native), English (IELTS 7.0, CET-6)